Lecture 08: free electron model; electrons in a lattice

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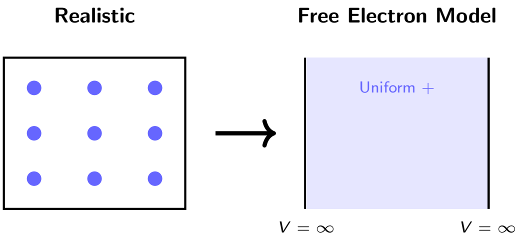
The free electron model: key assumptions

- Ignore electron-electron interactions (independent electron approximation)
- Replace ion lattice with uniform positive background charge (jellium model)
- Electrons are confined to a box with infinite potential walls at boundaries

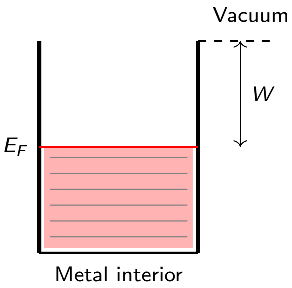


Why infinite potential well works

- Work function $W \sim$ 4–5 eV sets barrier height at metal surface
- Fermi energy $E_F \sim$ 5–10 eV means most electrons deeply bound
- Infinite wall approximation valid: $W \gg k_B T$ at room temperature

Thermal energy scale:

$$k_B T \approx 0.025 \text{ eV at } T = 300 \text{ K}$$



Particle in a 3D box: quantized energy

- Solve Schrödinger equation with $\psi = 0$ at walls (boundary conditions)
- Standing wave solutions require integer half-wavelengths fit in box
- Energy depends on three quantum numbers $n_x, n_y, n_z = 1, 2, 3, \ldots$

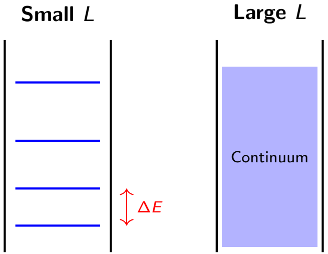
Free electron energy in cubic box

$$E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$$

Notation: h = Planck constant, m = electron mass, L = box side length

Why energy becomes quasi-continuous

- Level spacing $\Delta E \sim h^2/(8mL^2)$ shrinks as L increases
- For $L = 1$ cm: $\Delta E \sim 10^{-15}$ eV — far smaller than $k_B T$
- Discrete quantum levels merge into effective continuum at macroscopic scales



Energy level spacing

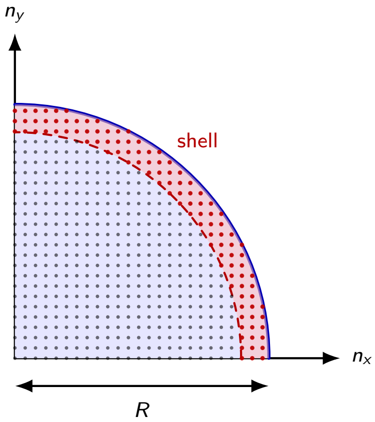
$$\Delta E \sim \frac{h^2}{8mL^2} \approx 10^{-15}\,\text{eV for } L = 1\,\text{cm}$$

Constant energy surfaces in n-space

- $E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2)$
- Define $R^2 = n_x^2 + n_y^2 + n_z^2$, so $E = \frac{h^2}{8mL^2} R^2$
- States with energy $\leq E$ lie inside sphere of radius R
- Only positive octant counts: $n_x, n_y, n_z > 0$

Radius in n-space

$$R = \sqrt{\frac{8mL^2 E}{h^2}}$$



Counting states: volume in n-space

- Total states with energy $\leq E$:

$$N(E) = \frac{1}{8} \times \frac{4\pi}{3} R^3 \times 2 \quad \text{(spin factor)}$$

- Substitute $R = \sqrt{8mL^2E/h^2}$
- Density of states $Z(E) = dN/dE$ gives states per unit energy

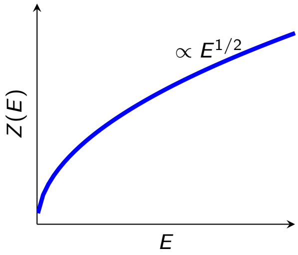
Integrated density of states

$$N(E) = \frac{\pi}{3} \left(\frac{8mL^2E}{h^2} \right)^{3/2}$$

Note: Factor of 2 accounts for spin-up and spin-down electrons (Pauli principle)

Density of states $Z(E) \propto E^{1/2}$

- Differentiate $N(E)$: $Z(E) = \frac{dN}{dE} = C \cdot E^{1/2}$
- Constant $C = \frac{4\pi L^3 (2m)^{3/2}}{h^3}$ depends on volume and mass
- Square-root dependence is signature of 3D free electrons



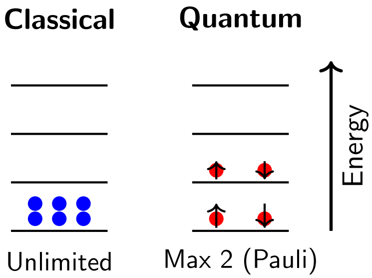
Density of states for free electrons

$$Z(E) = \frac{4\pi L^3 (2m)^{3/2}}{h^3} E^{1/2}$$

Why classical statistics fail for electrons

The problem:

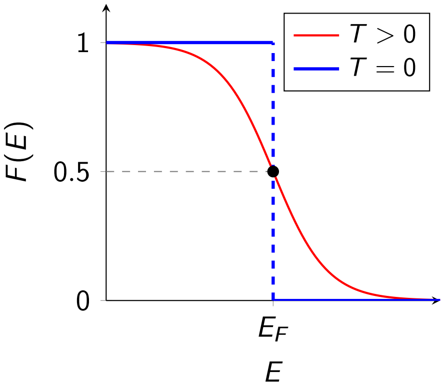
- **Classical Maxwell-Boltzmann:** any particle can occupy any energy state
- **Electrons are fermions:** Pauli exclusion forbids identical quantum states
- At metal densities ($\sim 10^{28}/\text{m}^3$), quantum effects dominate even at room T



The Fermi-Dirac distribution function

Key properties:

- $F(E)$ = probability state at energy E is occupied
- At $E = E_F$: occupation probability is exactly $\frac{1}{2}$
- Interpolates between fully occupied ($E \ll E_F$) and empty ($E \gg E_F$)



Fermi-Dirac distribution

$$F(E) = \frac{1}{\exp\left(\frac{E - E_F}{k_B T}\right) + 1}$$

Fermi-Dirac at $T = 0$: the sharp cutoff

Zero temperature limit:

- At $T = 0$: all states below E_F are filled, all above are empty
- $F(E)$ becomes a step function
- Electrons fill lowest available states like water filling a container

$T = 0$ limit

$$F(E)\big|_{T=0} = \begin{cases} 1, & E < E_F \\ 0, & E > E_F \end{cases}$$

Physical picture: Pauli exclusion forces electrons to stack up in energy, filling states from bottom up until all N electrons are accommodated.

Calculating E_F: counting electrons

Strategy: Total electrons = integral of occupied states

- At $T = 0$: integrate $Z(E)$ from 0 to E_F since $F(E) = 1$ below Fermi level
- Substitute $Z(E) = C \cdot E^{1/2}$ and solve for E_F

Total number of electrons N

$$
\begin{aligned}
N &= \int_0^{E_F} Z(E)\, dE = C \int_0^{E_F} E^{1/2}\, dE \\
&= C \cdot \frac{E^{3/2}}{3/2}\bigg|_0^{E_F} = \frac{2C}{3} E_F^{3/2}
\end{aligned}
\tag{1}
$$

Notation: $C = \dfrac{4\pi V (2m)^{3/2}}{h^3}$ includes factor of 2 for spin

Fermi level depends only on electron density

Solving for E_F:

- Invert $N = \frac{2C}{3} E_F^{3/2}$ to find E_F
- Result depends on N/V (electron density) only
- No temperature dependence at $T = 0$; weak T-dependence at finite T

Fermi Level

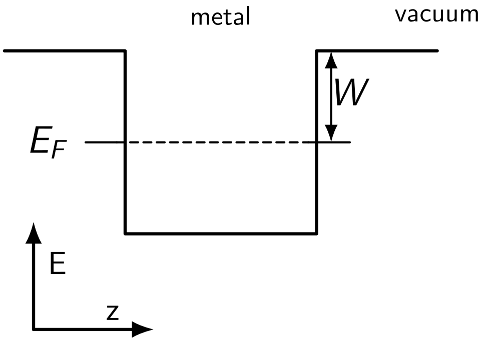
$$E_F = \frac{\hbar^2}{2m} \left(3\pi^2 \frac{N}{V} \right)^{2/3}$$

Typical values for metals:

- $E_F \sim$ 5–10 eV
- Fermi temperature: $T_F = E_F / k_B \sim 50{,}000$ K $\gg$ room temperature

Work function and thermionic emission

- Electrons at high T gain enough thermal energy to escape metal surface
- Only electrons with $E > E_F + W$ can overcome work function barrier
- Emission current depends on **tail** of Fermi-Dirac distribution above barrier
- Circuit must replenish emitted electrons for steady current



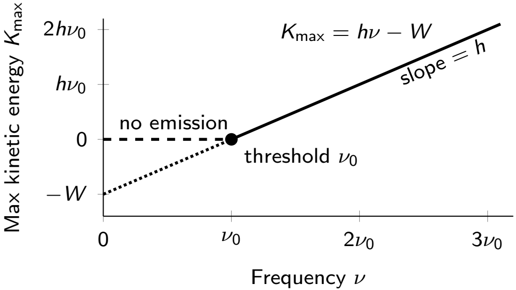
Photoelectric effect

Classical wave theory predicts:

- Any frequency works if intensity is high enough
- Energy accumulates continuously from wave

Experiment shows:

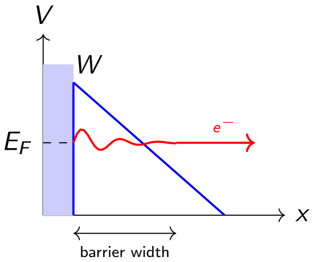
- Threshold frequency ν_0 exists regardless of intensity
- Electron *KE* depends on frequency, *not* intensity



Wave picture cannot explain threshold or frequency dependence!

Field emission: cold cathode tunneling

- At very high fields ($\sim 10^9$ V/m), electrons **tunnel** through barrier
- **Temperature-independent:** quantum tunneling replaces thermal activation
- Barrier width decreases with increasing field $\Rightarrow$ exponentially increasing tunneling



Applications:

- Field-emission microscopy
- Cold cathode displays
- Sharp tip electron sources

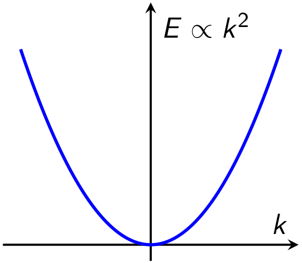
Recap: Free electron model and Fermi-Dirac statistics

What do you think are the main limitations of:

- The idea that electrons are free inside solids?
- The assumption that the electrons feel a "mean potential"?
- The approximations of a potential well?

Free-electron model: successes and assumptions

- Treats valence electrons as free particles in a uniform potential box
- Successfully explains metallic conductivity, thermionic emission, photoelectric effect, and more
- Assumes electrons experience no interaction with periodic lattice ions
- Predicts parabolic dispersion for all materials:



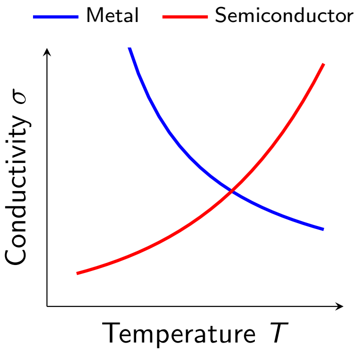
Free-electron dispersion

$$E = \frac{\hbar^2 k^2}{2m}$$

Critical failure: existence of insulators and semiconductors

- Free-electron model predicts **all** solids should conduct electricity
- It does not explain color or transparency
- Predicts wrong temperature dependence:

Metals: $\sigma \downarrow$ as $T \uparrow$
Semiconductors: $\sigma \uparrow$ as $T \uparrow$

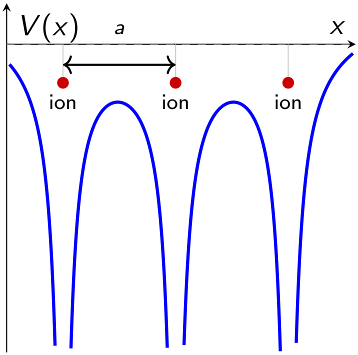


Introducing a periodic lattice potential

- Real crystals have periodic arrangement of ion cores
- Electrons experience attractive Coulomb wells at each lattice site
- Periodicity condition:

$$V(x + a) = V(x)$$

  where a is the lattice constant
- This periodic potential fundamentally modifies allowed electron energies



Problem

How do we solve the Schrödinger's equation for a periodic potential?

Dealing with periodicity: Bloch's theorem

Bloch wave function

For a periodic potential $V(x) = V(x + a)$, wave functions have the form:

$$\psi_k(x) = u_k(x)\, e^{ikx}$$

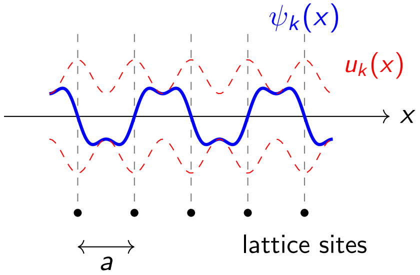
Notation:

- $u_k(x)$: periodic function with lattice period, $u_k(x) = u_k(x + a)$
- e^{ikx}: plane wave describing propagation
- k: wave vector (crystal momentum $= \hbar k$)

Key point: Wave vector k labels different quantum states in the crystal.

Physical meaning of Bloch waves

- Plane wave e^{ikx} describes propagation through crystal
- Periodic part $u_k(x)$ modulates amplitude near each atom
- Probability density $|\psi|^2$ is periodic (unit cells are equivalent)
- Bloch state means that electrons are not free nor localized



Bloch's theorem: key consequence

Translation property

Shifting by one lattice constant multiplies ψ by a phase factor:

$$\psi_k(x+a) = e^{ika}\,\psi_k(x)$$

Important consequences:

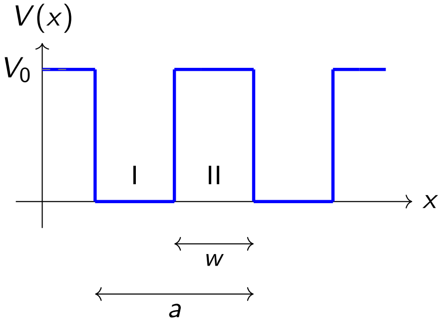
- Wave function is **not** periodic, but probability density **is**:

$$|\psi_k(x+a)|^2 = |e^{ika}|^2 |\psi_k(x)|^2 = |\psi_k(x)|^2$$

- All relevant physics is within the period $2\pi/a$ in k-space
- All relevant k is within the **first Brillouin zone**: $-\frac{\pi}{a} \leq k \leq \frac{\pi}{a}$

Kronig-Penney model: setup and assumptions

- Replace complex, periodic Coulomb potential with periodic square wells
- Well depth V_0, barrier width w
- Lattice constant: a
- Solve Schrödinger equation in each region, apply Bloch's theorem



Boundary condition

$$\psi(x+a) = e^{ika}\psi(x)$$

Wave functions in well and barrier regions

Notation: α = wave vector in well, β = decay constant in barrier, m = electron mass

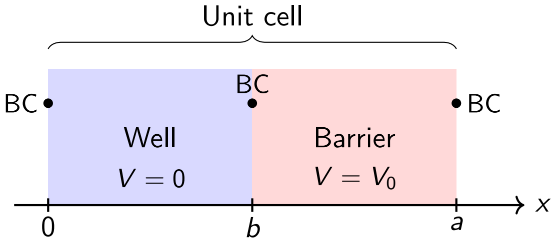
Solutions in each region

- In well, region I ($V = 0$): $\psi = Ae^{i\alpha x} + Be^{-i\alpha x}$
- In barrier, region II ($V = V_0$): $\psi = Ce^{\beta x} + De^{-\beta x}$

$$\alpha = \sqrt{\frac{2mE}{\hbar^2}}, \quad \beta = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \tag{2}$$

- Four unknowns (A, B, C, D) require four boundary conditions
- Continuity of ψ and $d\psi/dx$ at each interface

Applying boundary conditions (BC) considering periodicity



- Match ψ and $d\psi/dx$ at $x = b$ (well-barrier interface)
- Match ψ and $d\psi/dx$ at $x = a$ (barrier-well interface)
- Bloch condition: $\psi(a) = e^{ika}\psi(0)$ relates solutions across unit cell
- Nontrivial solution (i.e., $\psi = 0$ is not useful)

The Kronig-Penney dispersion relation

Notation: w = barrier width, a = lattice parameter

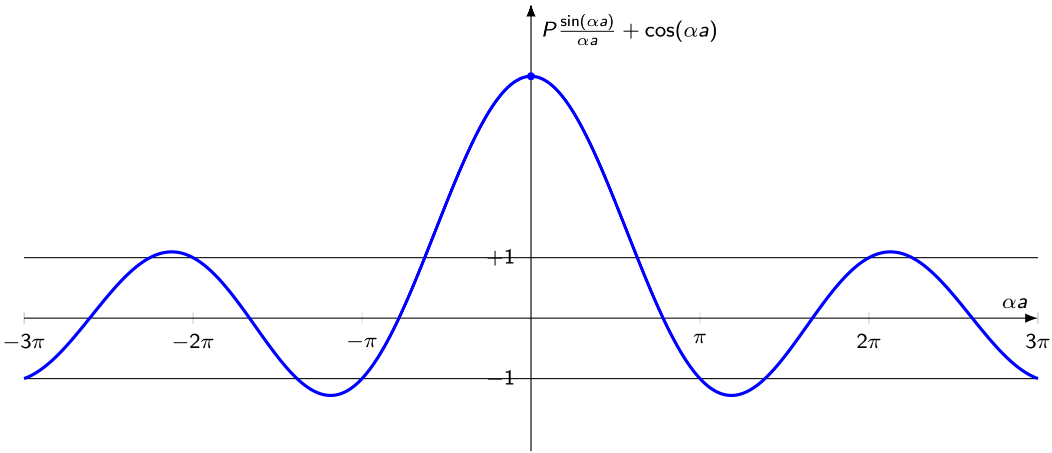
Kronig-Penney Equation

$$\cos(ka) = P\frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) \tag{3}$$

where $P = \frac{ma}{\hbar^2}V_0 w$ and $\alpha = \sqrt{\frac{2mE}{\hbar^2}}$

- Left-hand side ($\cos ka$) is bounded: $-1 \leq \cos(ka) \leq +1$
- Right-hand side depends on the geometry of the system, V_0, and electron energy E
- Solutions exist only when $P\frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) \leq 1$: **allowed energy bands**

Band gaps emerge at $k = n\pi/a$



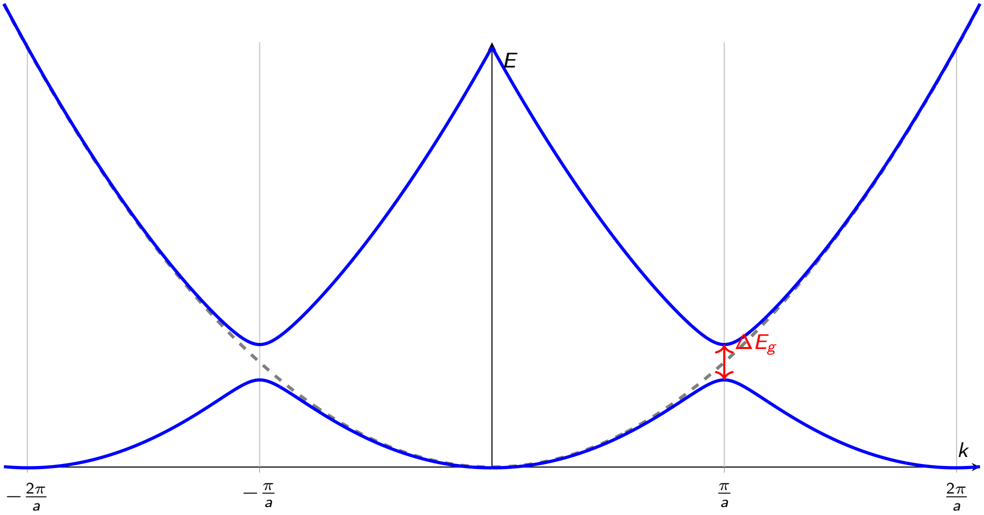
Constraints for the Kronig-Penney model

- $|\cos(ka)| > 1$: no real $k \rightarrow$ **forbidden regions**
- Gaps happen at Brillouin zone boundaries:

$$k = \frac{n\pi}{a}, \quad n = \pm 1, \pm 2, \ldots$$

- Gaps happen because of the Bragg condition: $2a = n\lambda$
- Electrons with the same periodicity of the lattice form standing waves with split energies

Gap opening in the Kronig-Penney model



Checkpoint: Kronig-Penney model summary

Conceptual questions:

- **Q:** Why use Bloch's theorem instead of simple periodic boundary conditions?
- **Q:** What determines whether energy E is allowed or forbidden?
- **Q:** Why do gaps appear specifically at $k = n\pi/a$?
- **Q:** Why did we draw multiple lines per k value on the last slide?
- **Q:** Why is there a parabola in dashed lines?

Key insight: Periodic potential + quantum mechanics $\rightarrow$ discrete allowed bands

From wave packets to particle dynamics

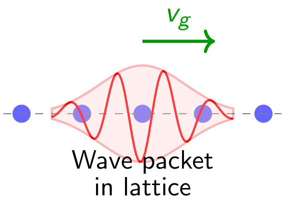
- Electrons in crystals are **wave packets** with group velocity

$$v_g = \frac{1}{\hbar}\frac{\partial E}{\partial k}$$

- Applied electric field changes crystal momentum:

$$\hbar\frac{dk}{dt} = -eE$$

- Goal: Connect quantum band structure to classical $F = ma$ dynamics



Deriving effective mass: step 1 – acceleration

Goal: Find acceleration of electron wave packet

Step 1: Acceleration is time derivative of group velocity

$$a = \frac{dv_g}{dt}$$

Step 2: Apply chain rule using $v_g = \frac{1}{\hbar}\frac{\partial E}{\partial k}$

$$a = \frac{d}{dt}\left(\frac{1}{\hbar}\frac{\partial E}{\partial k}\right) = \frac{1}{\hbar}\frac{\partial^2 E}{\partial k^2}\frac{dk}{dt}$$

Step 3: Substitute $\hbar\frac{dk}{dt} = F$ (applied force)

$$\boxed{a = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k^2}F} \tag{4}$$

Deriving effective mass: step 2 – definition

Compare with Newton's second law:

$$\text{Classical:} \quad a = \frac{F}{m^*} \tag{5}$$

$$\text{Derived:} \quad a = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k^2}F \tag{6}$$

Matching coefficients:

$$\boxed{m^* = \hbar^2\left(\frac{\partial^2 E}{\partial k^2}\right)^{-1}} \tag{7}$$

- Effective mass emerges naturally from **band curvature**
- m^* replaces free electron mass m in Newton's second law
- Quantum mechanics $\rightarrow$ classical dynamics with modified mass